BAHIRDAR INSTITUTION OF TECHNOLOGY

DEPARTMENT OF IS

OS INDIVIDUAL Project

Project title:

Arch Linux Installation ON VirtualBox

NAME : Mogessie Takele

ID:BDU1602173

Submitted to:LEC ALEMITU

Submitted date :12/05/2018

1. Introduction

Background

An Operating System (OS) is the core software that manages computer hardware and provides services to users and applications. Modern computing environments increasingly rely on **virtualization technology**, which allows multiple operating systems to run on a single physical machine safely and efficiently.

Motivation

Installing operating systems directly on physical hardware may risk data loss and hardware incompatibility. Virtualization provides a **safe, controlled, and flexible environment** for learning, testing, and experimentation.
This project focuses on installing **Arch Linux** in a **virtual machine** using **Oracle VM VirtualBox**, ensuring the host Windows operating system remains unchanged.

2. Objectives

2.1 Objectives of Knowing Virtual Installation of Operating Systems

- To demonstrate understanding of OS installation procedures in a virtual environment.
- To apply theoretical OS concepts such as boot process, filesystems, and partitioning.
- To gain hands-on experience with Linux OS without additional hardware.
- To develop troubleshooting skills related to OS installation and configuration.
- To explore OS resource management (CPU, memory, storage) using virtual machines.
- To understand virtualization architecture and its importance in modern computing.
- To prepare learners for real-world system administration tasks.

2.2 Objectives of Knowing Worldwide PC Brands

- To identify major PC brands capable of running multiple operating systems.
- To understand hardware diversity and OS compatibility.
- To relate OS performance with hardware specifications.
- To appreciate industry standards affecting OS design.
- To recognize CPU architectures (Intel, AMD, ARM) used globally.

2.3 Objectives of Knowing Worldwide Mobile Phone Brands

- To identify major smartphone brands and their operating systems.

- To understand the relationship between mobile hardware and OS design.
- To analyze mobile OS ecosystems (Android, iOS).
- To understand Android fragmentation and OS portability.
- To relate hardware innovation to OS evolution.

3. Requirements

3.1 Hardware Requirements

- PC/Laptop (HP ZBook)
- Processor: Intel with Virtualization support
- RAM: Minimum 4 GB (Recommended 8 GB)
- Storage: At least 40 GB free space

3.2 Software Requirements

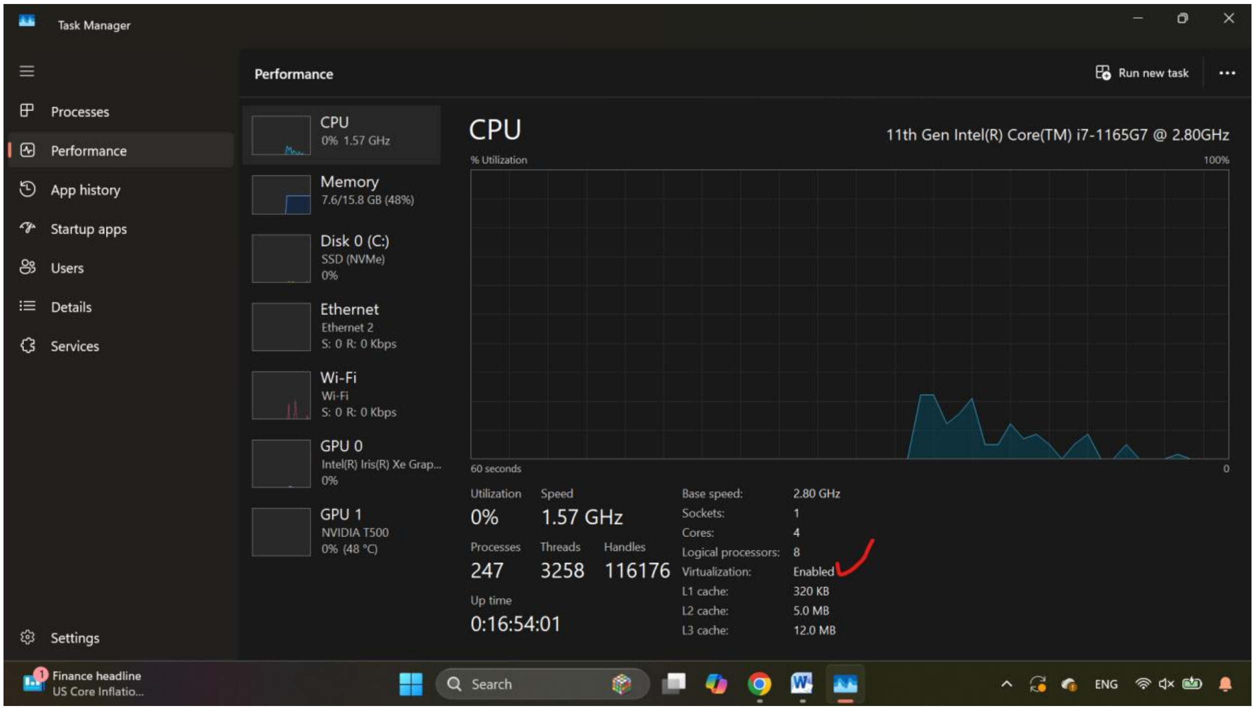
- Host OS: Windows 10/11
- Oracle VM VirtualBox
- Arch Linux ISO image
- Internet connection

4. Virtualization Tools Used

- **Oracle VM VirtualBox**
- (Other examples: VMware Workstation, Hyper-V)

5. Enable Virtualization (BIOS)

1. Restart the computer
2. Enter BIOS (F10 / F2 / Esc on HP)
3. Enable:
    - **Intel VT-x / AMD-V**
4. Save and exit

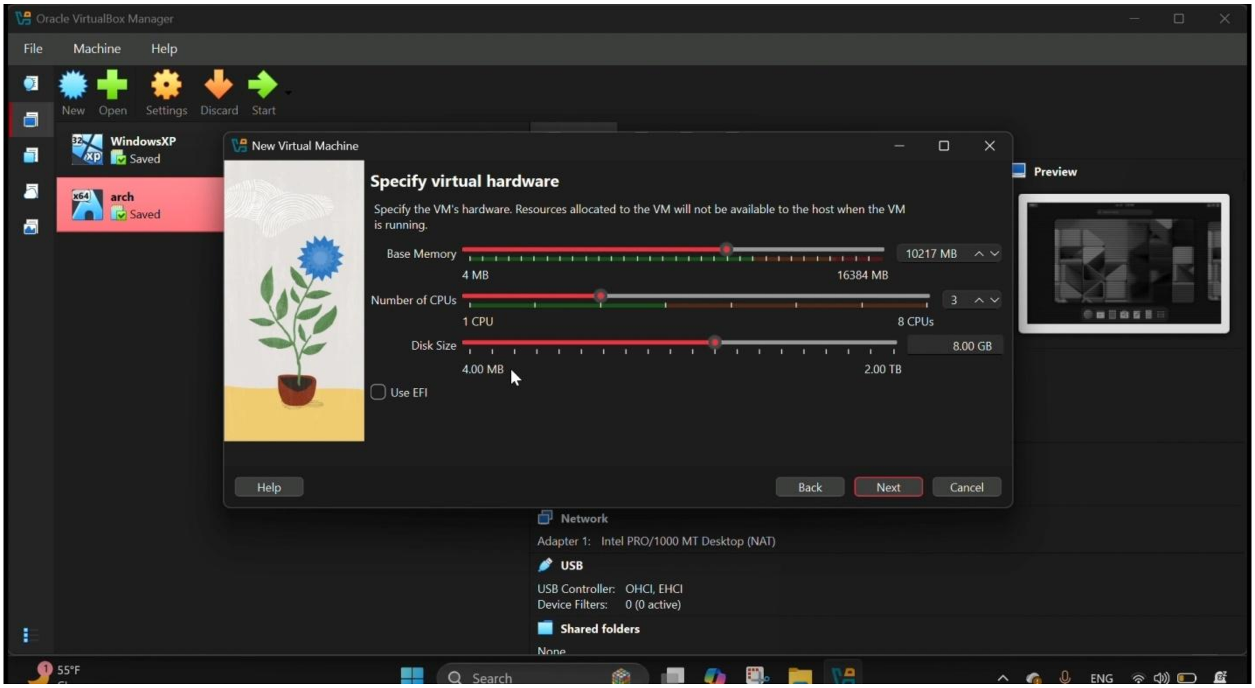


Without virtualization, VirtualBox will not run 64-bit systems.

6. Installation of Operating System in Virtual Environment

6.1 Virtual Machine Creation

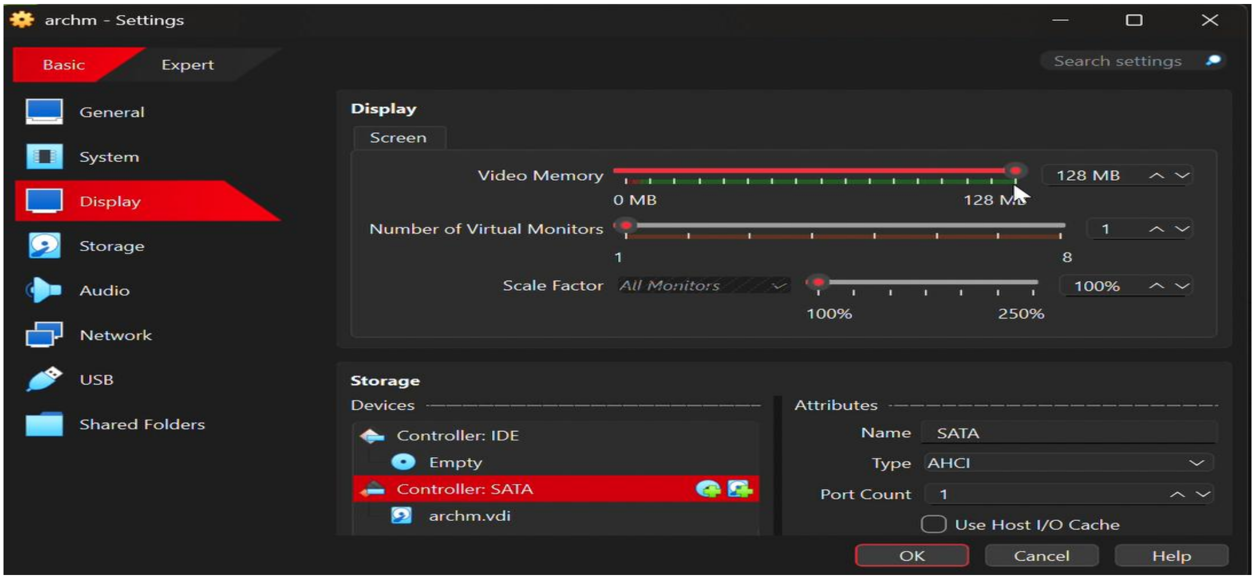
1. Open **Oracle VM VirtualBox**
2. Click **New**
3. Name: *Use your full name* (e.g., *Mogessie_Takele_ArchLinux*)
4. Type: Linux
5. Version: Arch Linux (64-bit)
6. RAM: 4096 MB
7. Disk: VDI → Dynamically allocated → 40 GB



1. Finish

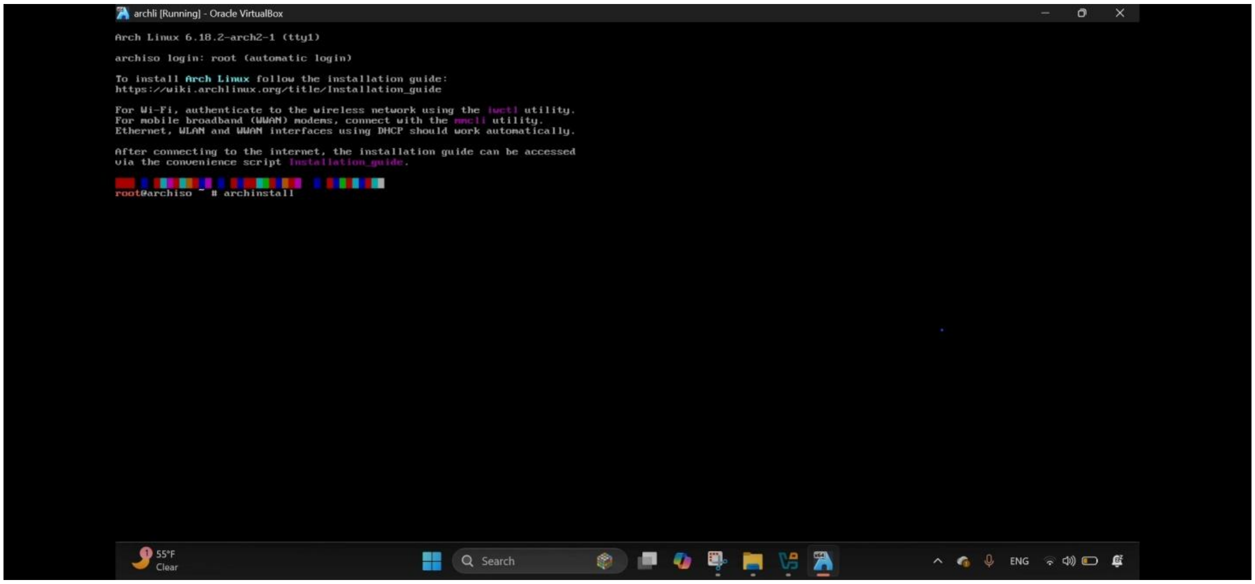
6.2. Attach Arch Linux ISO

1. VM → **Settings** → **Storage=120MB**
2. Controller IDE → Empty
3. Choose **Arch Linux ISO**
4. OK



7. Boot Arch Linux

1. Start VM
2. `archiso login: root`
3. Login automatically as root
4. Start installer:
5. `Archinstall`



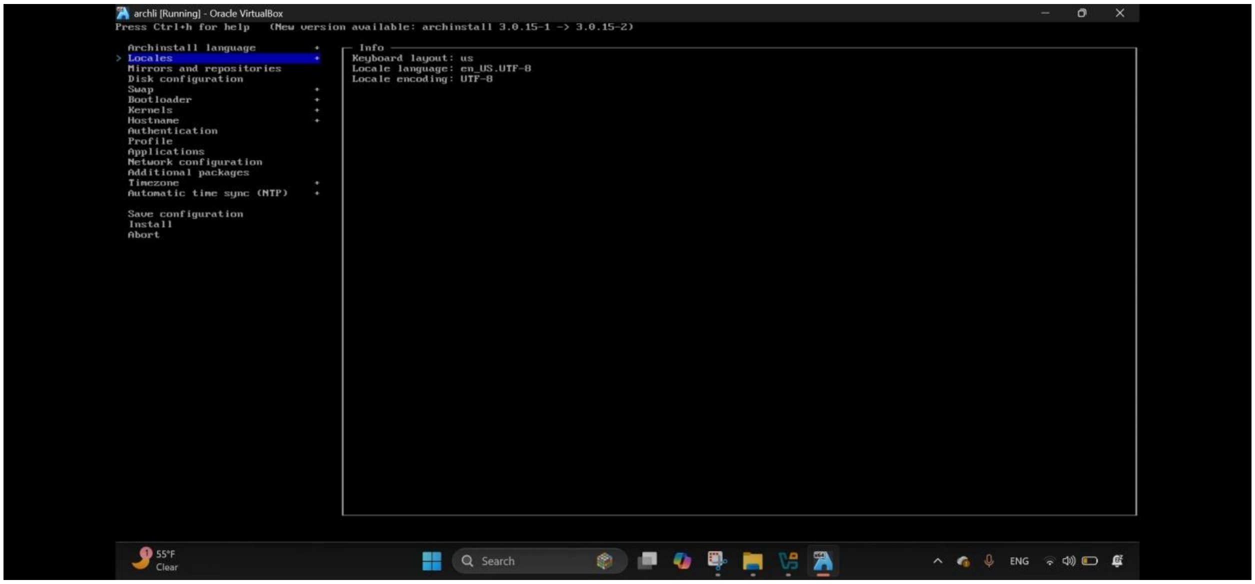
---

7. Archinstall Configuration

Archinstall Language

- Language: English
- Locale: en_US.UTF-8
- Keyboard: us

Locales



Mirrors and repositories

Select region=Germany

Disk Configuration

- **Use best-effort default partition layout**
- Disk: `/dev/sda`
- Wipe disk: **Yes**
- Partitions created automatically:

Mount	**Size**	**FS**
/boot	1 GiB	FAT32
/	Remaining	ext4

- Encryption: **Disabled**

Only the virtual disk is affected. Windows is safe.

Swap

- Swap on zram: **Disabled**

Bootloader

- **GRUB**

- Installed on `/dev/sda`

Kernel

- **linux (default)**

Hostname

- Example: `archlinux`

Authentication

- Root password: **Set**
- User created:
  - Username: `mogesie`

This step is mandatory. Installation will not proceed without it.

Profile

- Environment type: **desktop invironment**
- Selected: **GNOME**
- Packages installed include:
  - Gnome
  - Gnome-tweaks

Network Configuration

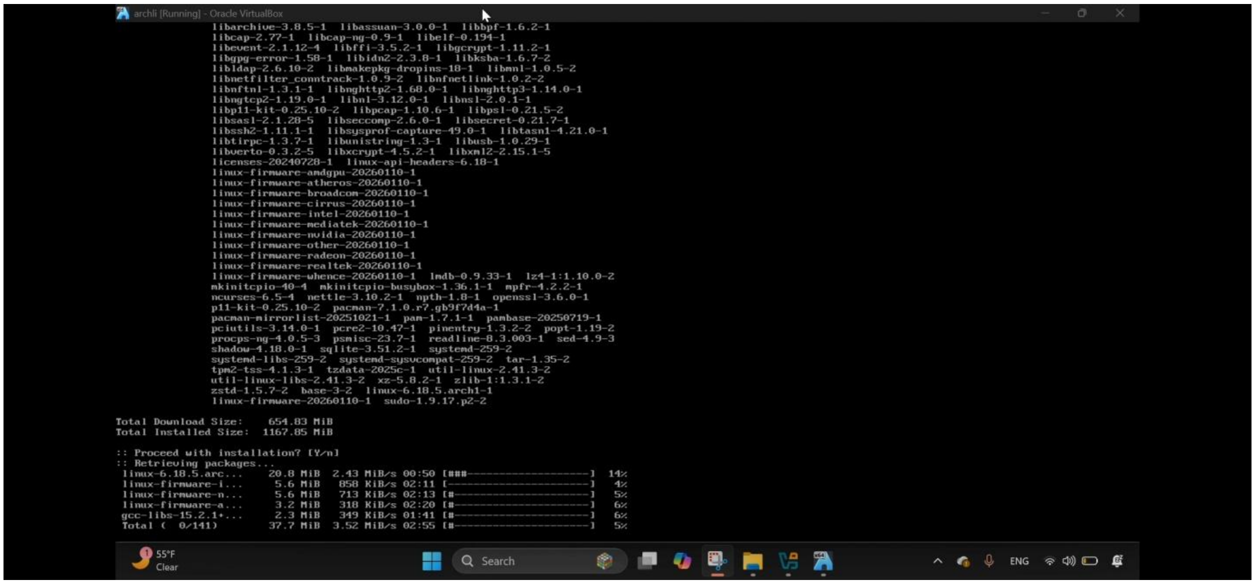
- **Use NetworkManager (iwd backend)**

Timezone & Time

- Region: **Africa**
- City: **Addis_Ababa**
- NTP: **Enabled**

8. Installation

1. Select **Install**
2. Confirm with **Yes**
3. System installs packages (10–20 minutes) based on internet connection



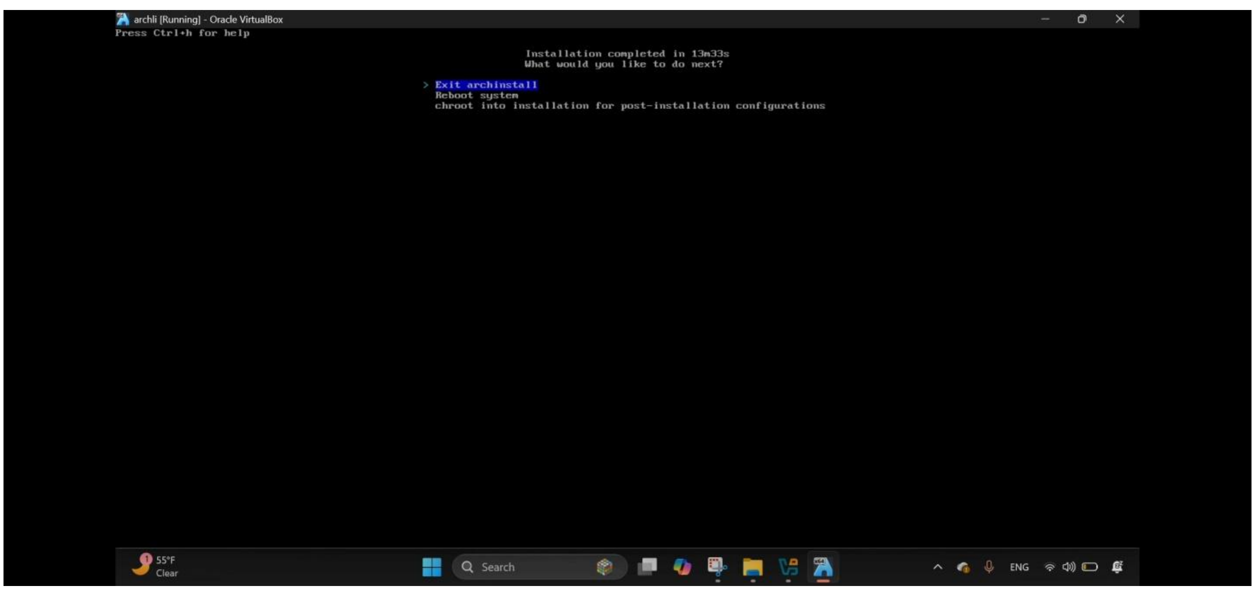
9. Installation Completed

Message shown:

```
Installation completed
```

Options:

- Exit archinstall
- Reboot system
- Chroot into installation



✔ Correct choice:

Reboot system

10. Important Reboot Step

Before reboot:

- VirtualBox → Devices → Optical Drives → **Remove ISO**

Then reboot.

11. First Boot (Installed System)

```
archlinux login:
```

Login:

```
username: mogesie
password: ****
```

12. Issues (Problems Faced)

1. **Network unreachable**
2. **startx command not found**
3. **Pacman mirror errors**
4. **GUI not starting**
5. **Incorrect profile selection**

Problem	Solution
Network unreachable	Enable NetworkManager
startx not found	Install Xorg and Desktop Environment
Pacman errors	Update mirror list
GUI not loading	Install correct desktop profile
Login errors	Verify username/password

☐ **`archinstall: ModuleNotFoundError`**

Reason:
`archinstall` only exists in the live ISO.

 Fix:
Remove ISO before reboot.

I try to enter `startx: command not found`

Reason:
xorg-xinit missing or `.xinitrc` not configured.

 Fix:

```
archinstall
```

Network unreachable

Reason:
NetworkManager not running.

 Fix:

- **Use NetworkManager (iwd backend)**

Check:

```
ping archlinux.org
```

13. Filesystem Support

Supported Filesystems

Filesystem	OS Support	Reason
NTFS	Windows	Large files, journaling
FAT32	Universal	Boot partitions
exFAT	Windows/Linux	Flash storage
ext4	Linux	Stability, performance
Btrfs	Linux	Snapshots
ZFS	Servers	Data integrity
APFS	macOS	Apple ecosystem

Chosen Filesystem: ext4

Why?

- Stable and reliable
- Best supported in Linux
- Low overhead
- Ideal for beginners

14. Advantages and Disadvantages

Advantages

- Safe installation without data loss
- Easy rollback and testing
- Supports learning multiple OSs
- Cost-effective

Disadvantages

- Lower performance than physical installation
- Requires sufficient RAM/CPU
- Limited hardware access

---

15. Virtualization in Modern Operating Systems

What is Virtualization?

Virtualization is a technology that allows multiple operating systems to run simultaneously on one physical machine.

Why Virtualization?

- Resource optimization
- Safe testing environment
- Cost efficiency
- Cloud computing foundation

How Virtualization Works

- Hypervisor manages hardware
- Virtual machines share resources

- Each VM runs isolated OS

---

16. Conclusion

This project successfully demonstrated the installation of **Arch Linux in a virtual environment** using **Oracle VM VirtualBox**. It enhanced understanding of operating systems, filesystems, virtualization, and troubleshooting. Virtualization proves to be a powerful and essential technology in modern computing.

---

17. Future Outlook / Recommendations

- Learn advanced Arch Linux system administration
- Explore server deployment
- Practice shell scripting
- Study cloud virtualization platforms (AWS, Azure)
- Experiment with different filesystems (Btrfs, ZFS)

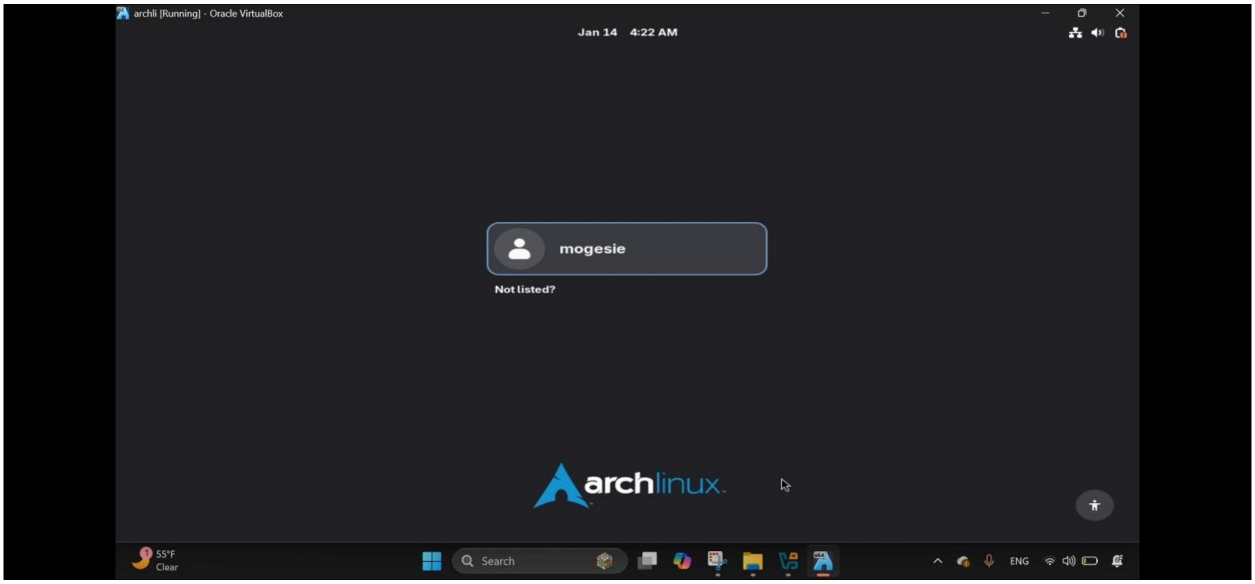
---

18. Evaluation Reflection

- ✔ Demonstrated OS installation expertise
- ✔ Clear understanding of virtualization
- ✔ Proper filesystem selection
- ✔ Practical troubleshooting skills
- ✔ Confidence in technical execution

19. Final Result

✔ Arch Linux installed inside VirtualBox
✔ Windows system untouched
✔ Network working
✔ GUI (GNOME WM) working
✔ Stable Arch system



Installing Arch Linux proves strong understanding of **Linux system administration**.

Safe Installation – Windows Not Affected